

Signal-Based vs. Equation-Based Modeling

Donnerstag, 24. Juli 2025 10:33

Comparison

Aspect	Signal-Based Modeling	Equation-Based Modeling
Causality	• Causality is explicit and fixed (causal) • Inputs and outputs are clearly defined for each subsystem	• Causality is implicit (acausal) • Direction of computation is not predefined; determined during model translation/compilation
Modeling Paradigm	• Procedural • direct signal-flow from component to component	• Declarative • Focuses on physical laws and conservation equations
Component Interfaces	• Ports represent signal in/out • Ports enforce the signal direction	• Ports represent physical connections • Ports are typically bidirectional
System Construction	• Connecting output and inputs ports of involved components manually	• System equations are automatically assembled from individual component equations and connection equations
Handling of constraints	• Constraints are handled manually or in procedural logic	• Can automatically • detect algebraic loops • add hidden constraints • perform index reduction
Mathematical basis	• Typically ODE or block-based algebraic simulation	• Models may include DAEs requiring structural analysis and transformation
Tool examples	Simulink, Simscape	Modelica
Simulation performance	• Often faster due to static dependency graphs	• Can be slower due to symbolic preprocessing and index handling
Event handling	• Based on signal thresholds • logical events	• Events arise from zero-crossing functions • Model-defined transitions between states (crossing functions, hybrid state)
Use	• Intuitive for control systems • Software defined logic	• Intuitive for physical systems (mechanics, thermodynamics, hydraulics)

Combination of Signal-Based and Equation-Based Modeling

- Complex mechatronic systems typically consist of
 - physical systems (mechanical, thermal, hydraulic)
 - logical control systems (controllers, sensors, software)
- Equation-based modeling is appropriate for describing physical interactions, constraints, and conservation laws
- Signal-based modeling allows the description of logical flow, control architectures, and time-driven processes
- Benefits of combining both:
 - Accurate and natural modeling of physical domains
 - Realistic feedback from sensors and actuators
 - Modularity and reusability
 - Clear interface between subsystems with different characteristics (e.g. FMU with fixed causality and equation-based models)

Relevant Concepts and Approaches for Hybrid, Heterogeneous System Modeling

Concepts	Purpose in Framework
Hybrid Modelling Support	• Enable connection of causal and acausal subsystems
Connector Interfaces	• Allow bidirectional connections for physical variables • Allow unidirectional ports for signal interfaces
Structural Analysis Tool	• Pantelide's Algorithm • Tarjan's Algorithm • Index reduction • Algebraic loop detection
Support for Co-Simulation	• FMU-based signal flow modules with explicit causality interact with acausal equation-based subsystems
Component Template Library	• Support reusable templates for equation-based and signal-based modeling elements

Solvers	• Symbolic solvers for acausal problem • Numeric solvers for procedural solvers
Dynamic Interface Mapping	• Allow automatic adaption between signal flow (causal) and equation-based (acausal) connections